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ORIGINAL INVESTIGATIONS

Return-to-Play for Elite Athletes With Genetic Heart Diseases Predisposing to Sudden Cardiac Death

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ABSTRACT

BACKGROUND People diagnosed with genetic heart diseases (GHDs) associated with sudden cardiac death (SCD) have historically been restricted from competitive sports. Recent data documenting return-to-play (RTP) experiences following shared decision making (SDM) suggest that cardiac event rates for athletes with a GHD are lower than previously described, thereby suggesting an opportunity to reconsider this paradigm.

OBJECTIVES The purpose of this study was to evaluate clinical outcomes among National Collegiate Athletic Association Division I university and professional athletes diagnosed with a GHD.

METHODS A multicenter retrospective analysis was performed to examine demographics, clinical characteristics, RTP outcomes, and cardiac events among elite athletes with a GHD.

RESULTS A total of 76 elite (66%, Division I, 34% professional) athletes (age 19.9 ± 5 years, 28% women) diagnosed with a GHD (hypertrophic cardiomyopathy [53%], long QT syndrome, long QT syndrome [26%]) comprise this cohort. Most athletes were asymptomatic (48 of 76, 63%) before diagnosis and had their GHD detected during routine preparticipation cardiovascular screening. Most athletes (55 of 76, 72%) were initially disqualified from their sport but subsequently opted for unrestricted RTP after comprehensive clinical evaluation and SDM. To date, (mean follow-up 7 ± 6 years), only 1 exercise-related (1.3%) and 2 nonexercise-related GHD-associated adverse cardiac events occurred. There have been no fatalities during follow-up.

CONCLUSIONS This is the first study describing the experience of athletes with a known SCD-predisposing GHD who are competing at the elite level. After careful evaluation, risk stratification, and tailoring of their GHD therapy, RTP following SDM appears associated with low, nonfatal events rates at elite levels of sport.
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ABBREVIATIONS AND ACRONYMS

AED = automatic external defibrillator

BCE = breakthrough cardiac event

GHD = genetic heart disease

HCM = hypertrophic cardiomyopathy

ICD = implantable cardioverter-defibrillator

LQTS = long QT syndrome

NCAA = National Collegiate Athletic Association

RTP = return to play

SCD = sudden cardiac death

VT = ventricular tachycardia

Sudden cardiac death (SCD) during exercise is the leading medical cause of death during exercise among young athletes. It is thought to be triggered by the inherent physiological stress that exercise places on the heart and often the tragic sentinel expression of a previously unrecognized underlying cardiac condition.[1,2] The exact incidence of SCD in young athletes is unknown and varies widely between reports. In an analysis among National Collegiate Athletic Association (NCAA) athletes, the overall incidence of SCD was 1:53,703 over a 10-year period but varied considerably by sex, race, and sport type.[1]

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Numerous data sets documenting underlying genetic heart disease (GHD) among SCD victims have led to consensus recommendations that advocate for immediate sport restriction following a GHD diagnosis.[3-5] GHD patients with type 1 long QT syndrome (LQTS), catecholaminergic polymorphic ventricular tachycardia, and hypertrophic cardiomyopathy (HCM) are currently recommended by U.S. guidelines to be restricted from sports, including those without markers of increased arrhythmic risk, because of concerns about increased risk of exercise-induced ventricular tachycardia (VT) and ventricular fibrillation.[6-8] Furthermore, recognition that exercise may accelerate disease progression and the risk of SCD among athletes with an arrhythmogenic cardiomyopathy (ACM) causative variants, particularly *PKP2*-mediated arrhythmogenic right ventricular cardiomyopathy,[9,10] has led to recommendations that such patients should be restricted from all competitive sports ranked higher than low-intensity Class 1A.[7]

However, emerging data suggest that the risks of competitive sport participation among many young athletes with GHD may be considered an unquantifiable and nonprohibitive risk. A recent study documented SCD and a nonlethal event rate of 1.16 per 100 athlete-years after risk stratification, evaluation, SDM, and return-to-play (RTP) in a cohort of athletes with LQTS.[11] In addition, a recent population-based study capturing 70,000,000 person-years of follow-up reported that SCD incidence during exercise was similar regardless of premortem vs postmortem diagnosis of HCM questioning the efficacy of exercise restriction for the prevention of SCD.[12] As a result, sports and genetic cardiology experts have gravitated toward a more patient-centered approach, utilizing the core tenets of shared decision making (SDM), to address RTP considerations among athletes diagnosed with GHD.[13-16] To date, data documenting the impact of this paradigm shift in the care of young athletes, particularly those who participate at elite levels of sport, are lacking.

RTP decisions at the elite level of sport are often publicly debated and highly scrutinized, because of the high visibility of the athletes and teams and widespread concerns about potential liability issues. However, an increasing number of young male and female elite athletes are electing to RTP despite these challenges. This level of competitive athletics historically disqualifies more athletes solely based on diagnosis than other levels, but there is no observational data reporting whether the decision protects the athletes from risk of sudden death. Accordingly, this multi-institutional study examined athletes with a SCD predisposing GHD competing at an elite, competitive level (Division I college or professional) and evaluated the outcomes of those allowed to RTP.

METHODS

Following Institutional Review Board approval, we collected and evaluated all elite athletes diagnosed with a sudden death predisposing GHD (ACM, catecholamine polymorphic ventricular tachycardia, dilated cardiomyopathy, HCM, idiopathic ventricular fibrillation, LQTS, left ventricular noncompaction-associated cardiomyopathy), then risk-stratified, treated, and given formal RTP approval at the 4 participating sports cardiology expertise centers: Windland Smith Rice Genetic Heart Rhythm Clinic (Mayo Clinic, Rochester), the Gagnon Cardiovascular Institute (Morristown Medical Center), the Cardiac Performance Program (Massachusetts General Hospital), and the Sanger Heart and Vascular Institute (Atrium Health) between July 2000 and July 2022.

For the purpose of this study, an "elite athlete" was defined as an individual who has played or is currently playing at an NCAA Division I (collegiate), Olympic, or professional (United States or International) level. For each athlete, electronic medical records were reviewed and information on demographics, duration of athletic career, history of present illness including cardiac symptoms or events before diagnosis, diagnostic testing (electrocardiogram, transthoracic echocardiogram, cardiac magnetic resonance, exercise testing, ambulatory monitor, and genetic testing), treatments were collected. The RTP process was then characterized by examining when they were disqualified, justification for initial disqualification, and how long each athlete waited until RTP approval.

As previously described[11,17] the sports cardiologist/genetic cardiologist at each participating institution followed a similar protocol utilizing SDM prior to facilitate RTP decision making, although the details may have varied from patient to patient because of the retrospective nature of this study and the absence of a standard, prospectively implemented SDM protocol. Briefly, each patient underwent an extensive consultation that included diagnostic disease confirmation, individualized guideline-directed treatment initiation, and disease-specific risk stratification at baseline and serially, including a discussion on risks with and without continued exercise. Each expert did not perform a uniform standardized SDM approach, because one universal approach is not known or appropriate given the heterogeneity of the diseases and sports involved.

Providers facilitated a discussion with the patient, their family members, and other important stakeholders as designated by the patient to address the potential risks, including SCD, of RTP. The overarching goal of this process was to ensure complete understanding of the risks and benefits of RTP and to arrive at a consensus about RTP among the athlete, their family, and the responsible physicians including designated team physicians. Once this discussion was completed, all additional potential stakeholders, including coaches, athletic trainers, and professional team or university administrators, were informed about the patient's RTP intentions, their individual risk of adverse outcomes, and how to appropriately manage their condition. In all cases, we confirmed that an automatic external defibrillator (AED) would be readily accessible during all training and competition. In isolated cases, patients chose to obtain their own AED.[18] All patients were routinely re-evaluated on an at least annual basis, often more frequently, by their designated care providers during which their risk profile and treatment strategy was reassessed.

The primary clinical outcome was breakthrough cardiac events (BCE) as defined by arrhythmic syncope or seizures, symptomatic nonsustained ventricular tachycardia, an appropriate VT and/or ventricular fibrillation-terminating therapy from an implantable cardioverter-defibrillator (ICD), sudden cardiac arrest with AED rescue, sustained VT, or SCD after diagnosis and initiation of treatment. These were divided further based on when they occurred: before RTP, after RTP at an elite level, or after conclusion of the athlete's (elite) athletic career. Additionally, the circumstances of the event with an emphasis on what the patient was doing at the time of the BCE (ie, during nonathletic activities of daily living vs during participation in their primary sport) were recorded.

TABLE 1 Demographics (N = 76)

Mean age at RTP, y	19.9 ± 5
Female	21 (28)
Race	
White	38 (50)
Black	28 (37)
Indian	1 (1)
Hispanic	1 (1)
Asian	1 (1)
Pacific Islander	1 (1)
Unknown/dot disclosed	4 (5)
Other	2 (3)
Diagnosis	
HCM	40 (53)
LQTS	20 (26)
ARVC	4 (5)
IVF	2 (3)
DCM	5 (7)
Other	5 (7)
Genotype positive	42 (55)
LQTS	20 (26)
HCM	16 (21)
ACM	4 (5)
DCM	2 (3)
Symptomatic before diagnosis	28 (37)
Syncope	10 (13)
Cardiac arrest	6 (8)
Palpitations/irregular heart rate	4 (5)
Sustained VT	2 (3)
Other	6 (8)

Values are mean ± SD or n (%).

ARVC = arrhythmogenic cardiomyopathy; DCM = dilated cardiomyopathy; HCM = hypertrophic cardiomyopathy; IVF = idiopathic ventricular fibrillation; LQTS = long QT syndrome; VT = ventricular tachycardia.

STATISTICAL ANALYSIS. Statistical analysis was performed using the PASW 18.0 software (SPSS). Results are expressed as mean ± SD for continuous variables or as number of cases (percentage) for categorical variables.

RESULTS

STUDY COHORT. The demographics of the study cohort are shown in Table 1. Among the 76 athletes included in this study, 50 of 76 (66%) were NCAA Division I athletes and 26 of 76 (34%) were professional athletes who are competing or have competed in the United States and/or internationally. The majority of athletes were men (55 of 76; 72%) with a mean age at RTP of 19.9 ± 5 years. Athletes self-identified one-half of the time (50%) as White and 37% as Black. The rest of the athletes self-identified as

Indian (1%), Hispanic (1%), Asian (1%), Pacific Islander (1%), other (3%), or unknown/undisclosed (5%). HCM represented the most common diagnosis (40 of 76; 53%) followed by LQTS (20 of 76; 26%). Additional diagnoses included dilated cardiomyopathy (5 of 76; 7%), ACM (4 of 76; 5%), idiopathic ventricular fibrillation (2; 3%), and other (5; 7%). Those categorized as "other" were diagnosed with various cardiomyopathies. Genetic testing revealed causal GHD variants in 42 of 76 (55%) of athletes. Approximately one-half of the athletes (40 of 76; 53%) were diagnosed with GHD because of abnormalities on preparticipation cardiovascular screening, whereas only 18 of 76 (24%) presented before diagnosis with cardiac symptoms, including syncope (9 of 18; 50%) and successful resuscitation from cardiac arrest (6 of 18; 33%) (Table 2) survived a sudden cardiac arrest event. Following initial cardiac GHD diagnosis, 55 of 76 (72%) athletes were disqualified from their sport by their initial provider, athletic director, or overseeing institution or organization. Following clinical evaluation, treatment initiation, and the SDM process at one of the institutions participating in this study, 69 of 76 (91%) athletes elected to RTP and were supported by their sports organization, 3 of 76 (4%) personally chose to withdraw from sport, and 4 of 76 (5%) remained disqualified by their primary provider or organization despite an expressed desire to RTP. The majority of those that were successful with RTP (55 of 76; 72%) returned to a sport with a high dynamic component and variable levels of static physiology (Figure 1).

FOLLOW-UP AND OUTCOMES. Patients were treated with a mix of pharmacological treatments (34 of 76, 45%), ICDs (24 of 76, 32%), and/or a left cardiac sympathetic denervation (6 of 76, 8%). The majority of ICDs (16 of 24, 66%) were placed for primary prevention ICDs based on guideline recommendations. During cumulative follow-up of 200 athlete-years, 1 of 76 (1.3%) athletes experienced an exertional BCE and 2 of 76 (2.6%) experienced a nonexertional BCE (Table 3). The first BCE was a male basketball player with HCM who presented with a nonexertional cardiac arrest. After initial disqualification by his team, he underwent secondary prevention ICD and was treated with a beta-blocker before returning to play. His BCE was an appropriate ICD shock while moving furniture. The second BCE occurred in a male hockey player who was diagnosed with LQTS after a syncopal episode. He was treated with a beta-blocker and returned to play the same year. Approximately 3 years later, he experienced 2 episodes of syncope with seizure-like activity. One of these events occurred during sport while he was coming off the bench and the other occurred at rest while he was eating "mac and cheese." Further adjustments to his medical management were performed and he returned to play without subsequent events. The third BCE occurred in a male hockey player with HCM who experienced a presumed nonarrhythmogenic syncopal episode. He was evaluated and returned to play untreated. He later experienced another exertional syncopal episode, after which he was started on a beta-blocker and subsequently returned to play with no further episodes during 52 months of follow-up.

TABLE 2 Evaluation of RTP

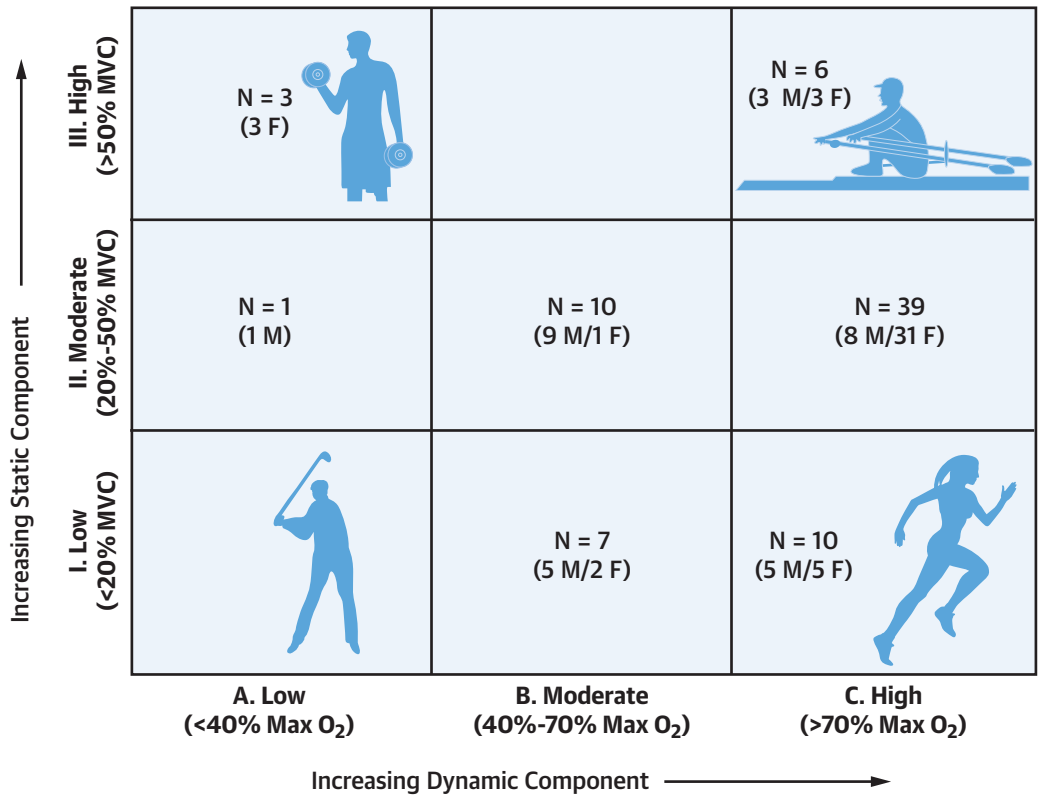
Evaluation of RTP	
Highest level of play	
Division I	50 (66)
Professional	26 (34)
Reason for diagnosis	
Abnormal cardiac evaluation	40 (53)
Syncope	9 (12)
Family history	7 (9)
Unrelated event	4 (5)
(Non)sustained VT	3 (4)
Cardiac arrest	6 (8)
Other symptoms	7 (9)
Initially disqualified	55 (72)
Chose to stop playing postdiagnosis	3 (4)
Remain disqualified	4 (5)
Follow-up	
Treatments while playing	
Pharmacological	34 (45)
ICD	24 (32)
Primary	16 (67)
Secondary	8 (33)
LCSD	6 (8)
BCEs	3 (4)

Values are n (%).
BCE = breakthrough cardiac event; ICD = implantable cardioverter-defibrillator; LCSD = left cardiac sympathetic denervation; RTP = return to play; VT = ventricular tachycardia.

DISCUSSION

To our knowledge, this is the first study to describe the experience of elite-level athletes who elect to RTP following a GHD diagnosis. Key findings are summarized graphically in the Central Illustration and described further in the following text. First, disqualification from sport immediately following GHD, irrespective of athlete preference, was common in this cohort. This finding confirms that conservative binary decision making remains pervasive among many community-based care providers. Second, RTP

FIGURE 1 Sports Participation for Return-to-Play Elite Athletes by Bethesda Categories



Distribution of sports participation of elite athletes in our study divided by sex and assigned to highest intensity sport (highest static and/or dynamic component). MVC = maximal voluntary contraction.

following consultation in a tertiary care center with both sports cardiology of GHD management expertise led to successful subsequent RTP. Importantly, the majority of athletes (55 of 76; 72%) resumed participation in a sport requiring high dynamic physiology. This finding underscores the feasibility of RTP among elite athletes with GHD who chose to engage in intense sporting disciplines following comprehensive risk stratification, individualized treatment, and SDM. This underscores the important relationship of the sports cardiologist/genetic cardiologist in the athlete medical team. Sports cardiology input regarding athlete risk and a plan of care is vital to the team physicians and organization/university because they are tasked with integrating the sports cardiology opinion into a final decision. A well-planned and thoughtful RTP protocol from your sports cardiology experts is an integral part of the sports medicine team for cardiovascular player health and safety.

Finally, we observed a low exertional BCE event rate of 1.3%, 2.6% nonexertional BCE and no SCD, and no successfully resuscitated cardiac arrests. This important finding suggests a favorable risk profile among carefully managed elite athletes with GHD by experts in the field and ongoing surveillance risk stratification.

MEDICAL AND LEGAL FRAMEWORK FOR DECISION-MAKING IN COMPETITIVE ATHLETES. Unconditional disqualification from competitive sports among athletes with an SCD-predisposing GHD has been the standard of care for decades. The 36th Bethesda Conference and its subsequent updates provides consensus sport eligibility recommendations for athletes with cardiovascular disease.[19] While this consensus document acknowledged the broad heterogeneity in genotypic and phenotypic manifestations of GHD, it recommended universal disqualification of athletes with GHD, citing both an inability to risk-stratify

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TABLE 3 BCE Details

	Patient 1	Patient 2	Patient 3
Sex	Male	Male	Male
Diagnosis	HCM	LQTS	HCM
Reason for diagnosis	Cardiac arrest	Syncope	Syncope
Sport and level	Basketball	Hockey	Hockey
Age at diagnosis	20	21	22
Age at RTP	21	21	23
Symptoms before diagnosis	Yes Cardiac arrest	Yes Syncope	Yes Syncope
Initially disqualified	Yes	No	Yes
Treatment	ICD, beta blocker	Beta blocker	Beta blocker
BCE	ICD shock	Syncope	Syncope
Age at BCE, y	22	24	24
Circumstances of BCE	While moving furniture	While coming off the bench and making mac and cheese	While working out

BCE = breakthrough cardiac event; HCM = hypertrophic cardiomyopathy; ICD = implantable cardioverter-defibrillator; LQTS = long QT syndrome; RTP = return to play.

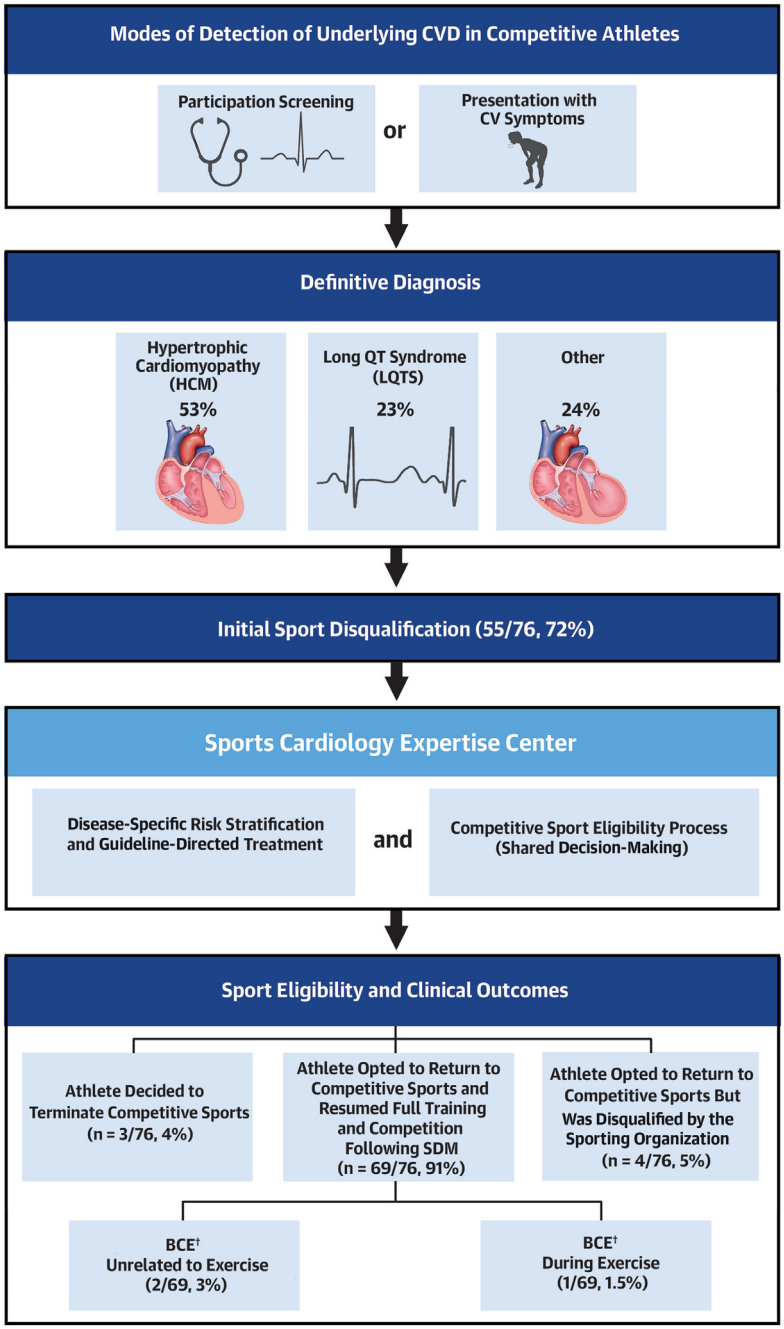
individuals and the absence of safety data.[19] This approach has been challenged in the past including the widely cited legal case of *Knapp vs. Northwestern University*. In 1995, Nick Knapp experienced a cardiac arrest during basketball practice and was diagnosed subsequently with HCM. A secondary prevention ICD was placed, and he was disqualified from future athletic participation. The athlete subsequently sued Northwestern University claiming a violation of his rights under section 504 of the Rehabilitation Act. Ultimately, the courts upheld the university's right to exclude the athlete from sport and ruled that sport eligibility decisions lie with the team physician. The court's ruling stipulated that eligibility decisions overseen by team physicians should be made *reasonably and rationally* by taking into consideration all relevant medical facts, opinions of expert medical consultants, and current (not prior) clinical practice guidelines. The team physician medical judgment model, still a very appropriate standard of care, prioritizes determination of the level of risk through consideration of sport type, age, sex, and individualized diagnostic data. Increasingly, the team physician model integrates SDM, a practice that has become standard of care across all disciplines that engage in complicated medical decision making. We anticipate that our data will assist team physicians and their cardiology consultants with this challenging task. In addition, and in stark contrast to what has previously been proposed,[20] our experience demonstrates that some athletes elect to self-disqualify after SDM. This important observation renders the notion that all athletes should be considered a "limited control" population both antiquated and inaccurate.

THE ROLE OF SDM. Carefully evaluated and optimally treated people with GHD, including competitive athletes at sub-elite levels of sport, can exercise with a low risk of cardiac events. In addition and in stark contrast to what has previously been proposed,[20] our experience demonstrates that some athletes elect to self-disqualify after SDM.[21-23] This study provides similar and confirmatory data derived from elite athletes participating at the NCAA D1 or professional levels of sport and confirms that some athletes with a GHD can safely RTP after comprehensive diagnostic evaluation, risk stratification, and treatment.[24,25] SDM, the process that integrates patient input with care providers medical opinions, represents a crucial step in this process. The management of each elite athlete in this cohort was guided by SDM, a shift from the historically paternalistic approach that has been pervasive among sports cardiologists and GHD experts. SDM emphasizes the values and personal preferences of the fully informed patient to ultimately arrive at an outcome satisfactory to all participants. The 2020 AHA/ACC HCM recommendation emphasized the role of SDM in all aspects of HCM care including decisions regarding sports eligibility.[26] Diverging from prior guidelines, current recommendations now endorse HCM patient participation in vigorous competitive sports after they have been fully informed by an expert provider about the potential risks (including sudden death). Findings from this study, specifically as they pertain to elite levels of sport, suggest that this new recommendation is both feasible and associated with low adverse event rates.

We also want to emphasize that SDM is a dynamic conversation between all stakeholders in the decision and each expert on this study engages in it differently. This is not an easy process or a boilerplate discussion and expectedly varies by provider and athlete scenario. A recent editorial by Kim and Dickert[27] discussed the challenges with the "how-to" part of SDM, which is not well vetted in this space and can be subject to important issues, including biases, differences in risk assessment, and how the risk is conveyed. Engaging in the SDM process does not guarantee RTP approval, but rather allows a discussion on risk stratification, treatment options, and next steps for all stakeholders. This data does not imply that all cardiologists should or will be able to have SDM conversations for this level of athlete and will likely be limited to experts with experience in both the particular disease encountered and the intricate nuances of RTP discussions.

IMPLICATIONS OF ATHLETIC DISQUALIFICATION. While this study was not designed to directly address

CENTRAL ILLUSTRATION **Return-to-Play Among Elite Athletes With Genetic CVD**



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A multicenter, retrospective review of 76 athletes (National Collegiate Athletic Association [NCAA] Division I/professional level) with a genetic heart disease (GHD) (53% hypertrophic cardiomyopathy [HCM], 26% long QT syndrome [LQTS]) and cleared for return-to-play (RTP) (28% women, mean age 19.9 ± 5 years). Most were asymptomatic (48 of 76, 63%) with their GHD detected during routine preparticipation screening. 55 of 76 athletes (72%) were initially disqualified. Most athletes (55 of 76, 72%) were initially disqualified from sport but subsequently opted for unrestricted RTP after comprehensive clinical evaluation and shared decision making (SDM). There were only 1 exercise-related (1.3%) and 2 nonexercise-related adverse cardiac events, and no deaths (mean follow-up 7 ± 6 years).
†BCE = breakthrough cardiac event, follow-up 7 ± 6 years.

the implications of sport disqualifications, several issues deserve attention. Most elite athletes work extremely hard over many years to reach the highest levels of sport. Psychological consequences of disqualification can include depression, anger, frustration, uncertainty about recreational exercise safety, and increased stress adjusting to a lifestyle without competitive athletics.[28-30] In extreme cases, suicidal tendencies may arise following disqualification. Social consequences may include isolation from both peers, including their primary friend group, which is often comprised of teammates, and important mentors including coaches. Financial consequences of disqualification may include loss of academic and athletic scholarship at the collegiate level and loss or decrease in salary and signing bonuses at the professional level. Finally, sport disqualification may lead to a sedentary lifestyle with its negative implications, including weight gain and the development of atherosclerotic risk factors resulting from poor mental health and overall wellbeing.

In summary, sport disqualification comes with a myriad of direct risks that should be considered throughout the SDM process.

ATHLETES AND PLAYING WITH AN ICD. While there remains some debate about the safety of playing competitive sports with an ICD, the 2021 guideline for patients with ICDs endorsed participation in competitive sports or intense recreational exercise after SDM that conveys the estimated risk and also includes coaches, schools, communities, or teams.[31] This recommendation is based on recent data documenting low event rates among athletes with ICDs who continue to play sports following device implantation.[32,33] Our data provides additional support for safety of ICDs in athletes and now extend this concept to the elite levels of sport including elite-level athletes. Specifically, there were 24 elite athletes in our cohort with ICDs, including 8 who had an ICD placed for secondary prevention, who returned to play after ICD placement. ICDs were placed based on physician judgment and guideline recommendations, and some were placed before coming to one of the expert centers for RTP consideration. These athletes have continued to play at an elite level under close clinical surveillance.

STUDY LIMITATIONS. First, we acknowledge that our cohort was comprised of athletes that largely self-referred to participating tertiary care sports cardiology/genetic cardiology centers. As such, the safety profile reported in this paper may be higher than that observed in alternative clinical settings. In addition, the low rate of athletes who elected to withdraw from sports following SDM may under-represent preference patterns of larger unselected groups of athletes. Second, we cannot exclude the possibility of survival or selection bias, because we had a relatively small number of athletes who presented following resuscitated SCD. It is possible that RTP preference and clinical outcomes may differ in this group. Third, there was relatively low representation of contact sports (ie, American-style football, hockey) in our cohort, and thus we cannot exclude the fact that outcomes among athletes participating in these disciplines may differ from our results.

Additionally, athletes competing at the elite level often have access to more resources than athletes at other levels. Included in the athlete's SDM/RTP counseling session is a transparent discussion that no evaluation process is perfect and that there is no such thing as zero-risk Therefore, athletes forecasted to be at low risk may still have cardiac events. Consequently, a necessary part of SDM must include confirmation of cardiac arrest mitigation planning and resources, which often includes having a personal AED. In addition, SDM in these scenarios is an ongoing process and requires the ability to provide follow-up and continuous risk stratification, which may be less feasible at some competitive levels. As a result, this underscores a limitation of the study, and to be generalizable to every athlete scenario, athletes and institutions must be diligent to maintain risk stratification and resources for player health and safety.

Finally, our retrospective study design and limited sample size do not permit definitive estimations of BCE incidence. Ongoing prospective work is being conducted to address many of these issues (the ORCCA [Outcomes Registry for Cardiac Conditions in Athletes] study[34]).

CONCLUSIONS

This is the first multicenter study to examine RTP outcomes among elite-level athletes with GHD. This study included athletes with sudden death predisposing GHD, the vast majority of whom have successfully returned to competitive sports at the NCAA Division I collegiate or professional level after a comprehensive clinical evaluation and the use of SDM. These data suggest that sport after a GHD diagnosis is both feasible and associated with low

adverse event rates, even in elite-level athletes. Although we anticipate that our data may assist practitioners charged with the care of young athletes with GHD, future larger-scale prospective data are warranted to fully define the risk and benefits of RTP following a GHD diagnosis.

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PERSPECTIVES

COMPETENCY IN PATIENT CARE AND PROCEDURAL SKILLS: Although historically, athletes with GHD predisposing to SCD were restricted from playing competitive athletics, those who return to play after expert risk stratification and tailoring of therapy face a low risk of cardiac events.

TRANSLATIONAL OUTLOOK: Further efforts are needed to improve access to centers with expertise in risk stratification and management of patients with GHD seeking to engage in competitive athletics at elite levels.

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